



Habib University

School of Science & Engineering

Course:	Linear Integrated Circuits
Semester	Fall 2025
Assignment #	2
Due Date :	October. 30, 2025
Instructor :	Syed Arsalan Jawed
Total Marks :	100

Instructions for Students

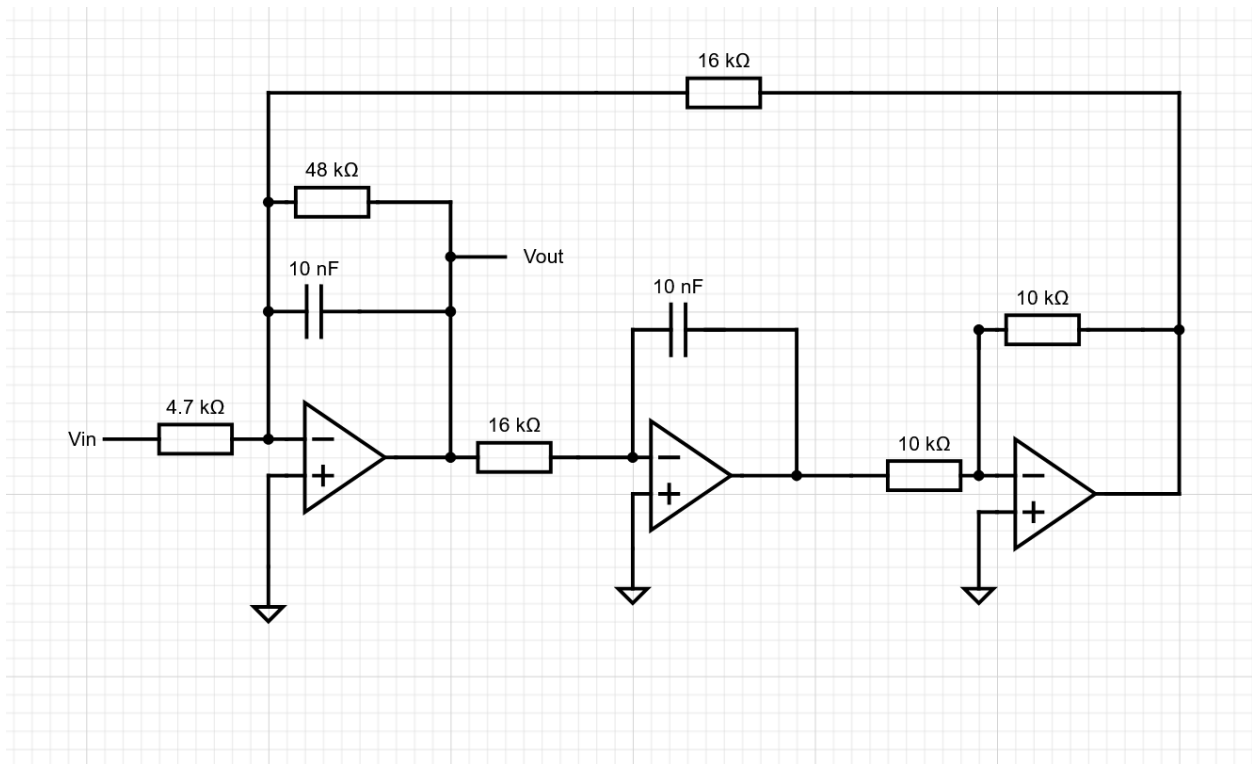
1. Print the following question paper and provide your answer within the empty spaces given after each question.
2. Keep your answers concise and to the point. DO NOT WRITE LENGTHY Descriptions. Provide neat and tidy calculations and plots.
3. Make reasonable assumptions related to concepts and associated parameters motivated by your text and reference book.
4. Support your answer with equations and simulation where asked and needed.

CLOs	Description	Learning-domain level
CLO - 1	Apply working principles and behavior of Opamp based linear integrated circuits to solve modern electronic circuits for various applications.	Cog – 3
CLO - 2	Analyze operational, differential, power amplifiers, Filters, Oscillators and Data Converters for the modern electronic applications.	Cog – 4
CLO - 3	Design/Simulate linear integrated circuits such as Filters, Oscillators, Data Converters.	Cog – 6

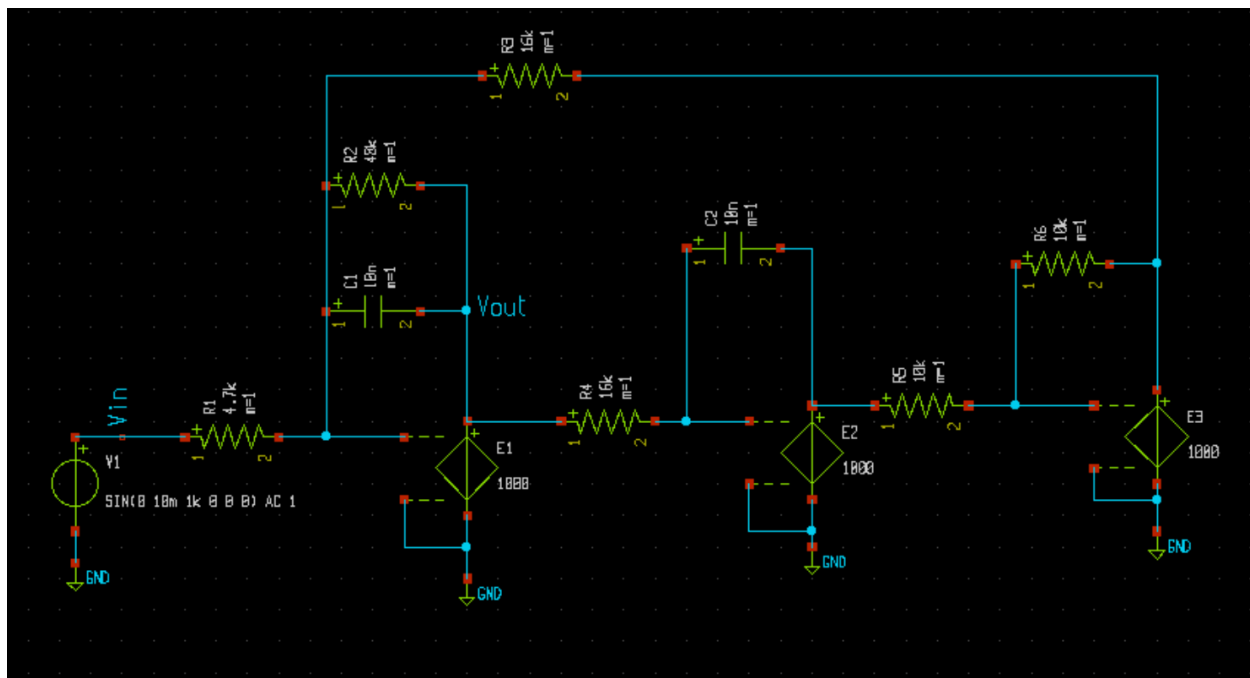
Question # 1 : [CLO-1, 2,3]

Design a bandpass Biquad filter using two integrator approach, this filter is required for audio amplification applications, achieve passband gain of >5 and stopband attenuation of <0.2 .

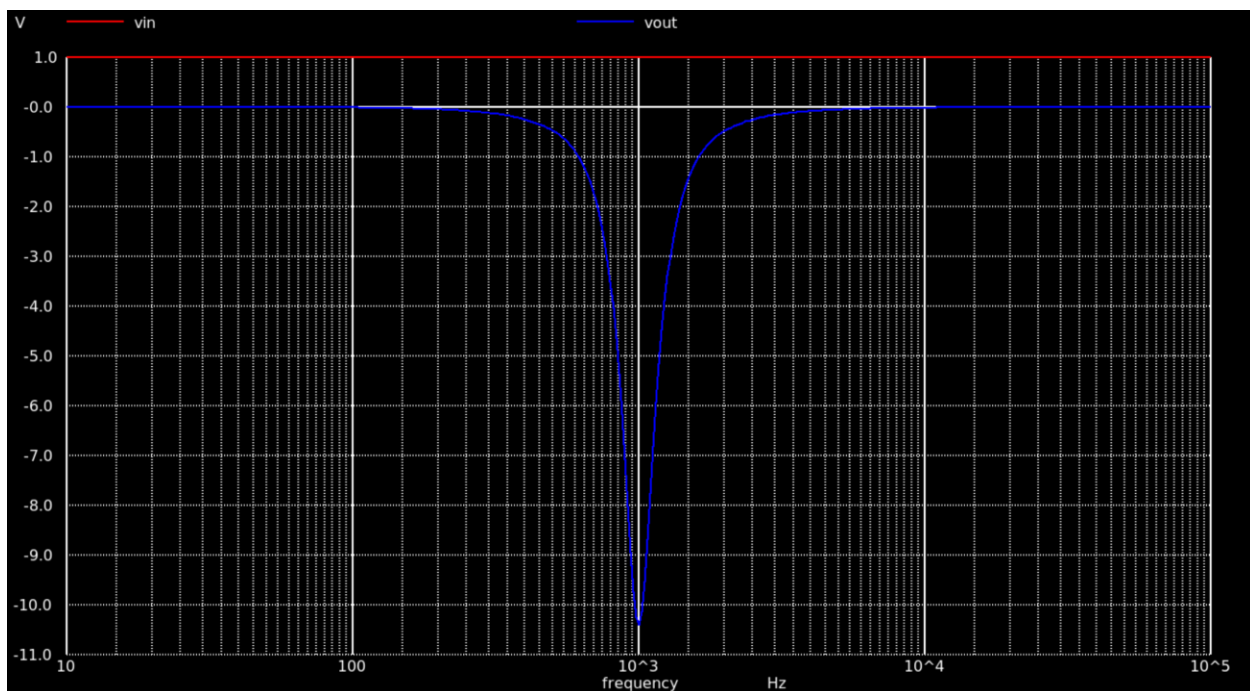
- 1) Draw the Opamp based RC network and calculate the required values of R_s and C_s . Submit this schematic.



- 2) Draw the circuit in XSCHEM and Simulate using NGSPICE using ideal Opamp

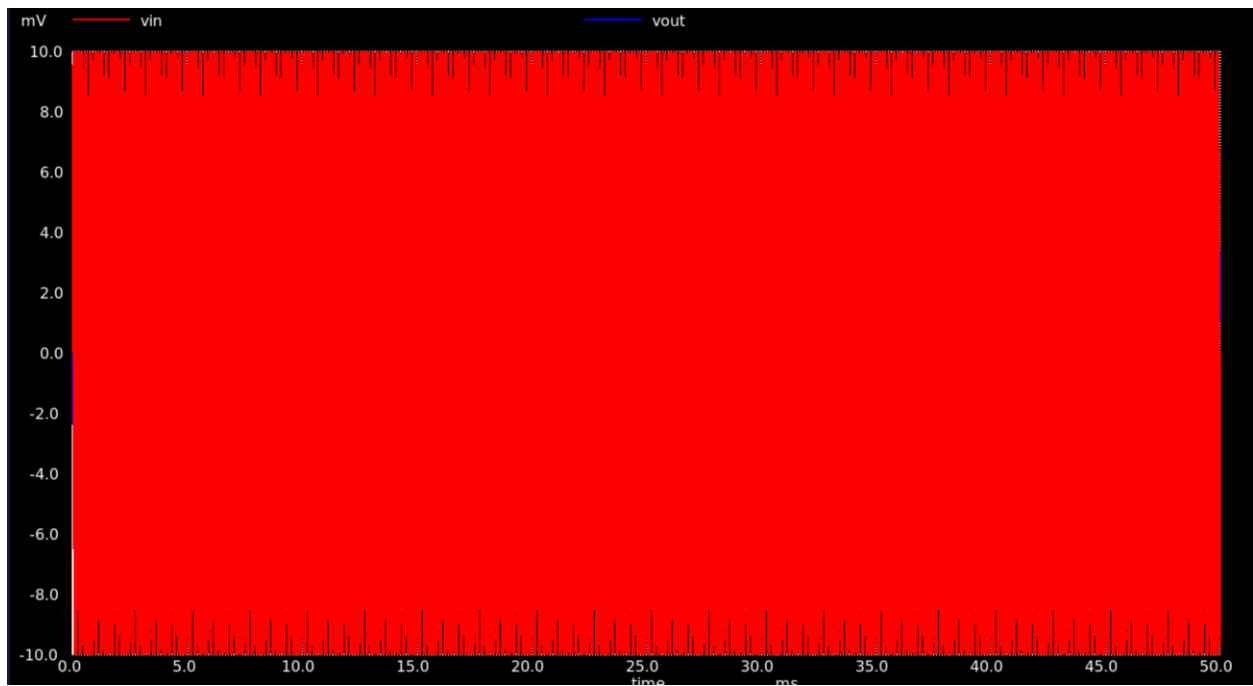


3) Run AC simulation to show the band of the filter.

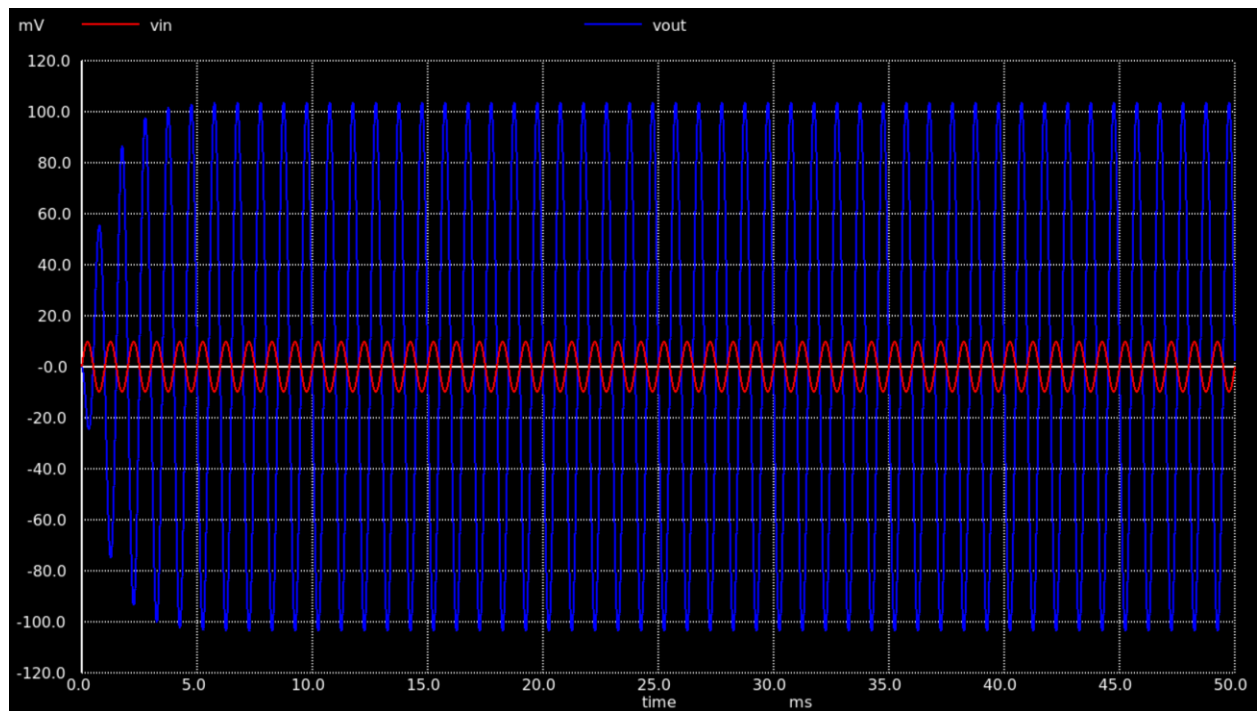


4) Run transient simulation to show how it treats in-band and out-of-band frequencies.

10k



1k



100

